

STANDARD OPERATING PROCEDURE

*Choosing a new technology the
company should start on/adopt*

REVISION HISTORY

Revision No. / Version No.	DATE	Prepared by:	Approved by:	Details of Change
1.0	18-Aug-2024	Anant Yende, CTO	Revati Kumar, CEO	

1. INTRODUCTION

This Standard Operating Procedure (SOP) outlines the steps to be carried out when there is a need to venture in to a new technological domain/programming language.

2. OBJECTIVES

This Standard Operating Procedure (SOP) aims to provide a comprehensive guide for evaluating and selecting new technologies within the organization. Given the constant advancements in the technological domain, it is essential to ensure that any new technology or programming language adopted by the organization undergoes a thorough due diligence process. This SOP is designed to mitigate potential roadblocks, cost overruns, and resource misallocation by providing a structured approach to technology selection.

3. DEFINITIONS

- Technology: Any software, programming language, framework, tool, or system that could be adopted for use within the organization.
- Due Diligence: A systematic process of evaluating the merits, risks, and implications of adopting a new technology.
- Stakeholders: Individuals or groups who have an interest in or are affected by the adoption of new technology.

4. RESPONSIBILITIES

- Technology Evaluation Team: Responsible for conducting the research, evaluation, and PoC development.
- CTO/Solution Architect: Oversees the overall process, ensuring that timelines are met and resources are allocated appropriately.
- CEO: Provides final approval for technology adoption and ensures alignment with organizational goals.
- Training Department: Responsible for developing and delivering training programs for the new technology.

5. PROCESS

1. Why we need a new technology ?- *Initiation of Technology Evaluation*

a) Identify the Need: Determine the business or technical need for a new technology. This could arise from a gap in current capabilities, performance issues, security considerations, or strategic business goals etc as detailed below but not limited to.

i. Business Drivers:

1. Strategic Goals: Align the need for new technology with the organization's strategic objectives. This could involve supporting business expansion, improving customer experience, enhancing operational efficiency, or enabling digital transformation.
2. Market Demand: Identify market trends or customer demands that the organization needs to respond to. This could include the need to offer new products or services, enter new markets, or meet changing customer expectations.
3. Competitive Pressure: Consider the need to adopt new technology to maintain or enhance the organization's competitive edge. This might involve adopting innovations that competitors are leveraging or addressing areas where the organization is falling behind.

ii. Technical Requirements:

1. Performance Improvements: Determine if there are performance-related challenges with current systems that necessitate the adoption of new technology. This might include improving speed, scalability, reliability, or reducing latency.
2. Security Enhancements: Identify if there are security concerns that the current technology stack cannot adequately address. The need for stronger encryption, better access controls, or more comprehensive compliance with industry regulations might drive the need for new technology.
3. Integration Needs: Assess whether the organization needs to improve integration capabilities with other systems, whether internal or external. The ability to seamlessly exchange data with customers, or other departments might necessitate new technology.

iii. Operational Challenges:

1. **Cost Reduction:** Evaluate whether there is a need to reduce operational costs, which might be achieved by adopting more efficient or automated technologies. This could include reducing licensing fees, minimizing manual processes, or optimizing resource usage.
2. **Process Optimization:** Identify inefficiencies in existing workflows or processes that could be streamlined through new technology. This could involve automating repetitive tasks, improving data accuracy, or enabling better decision-making through advanced analytics.
- iv. **Resource Constraints:** Determine if current resources are insufficient to meet organizational needs. The adoption of new technology might be necessary to better utilize resources, fill skill gaps, or manage increased workloads.
- v. **Regulatory and Compliance Needs:**
 1. **Legal Requirements:** Consider any regulatory changes or compliance requirements that necessitate the adoption of new technology. This might involve ensuring data privacy, meeting industry standards, or adhering to new laws.
 2. **Audit and Reporting Needs:** Identify if there is a need to improve audit trails, reporting capabilities, or documentation processes to comply with internal policies or external regulations.
- vi. **Innovation and Growth Opportunities:**
 1. **Innovation Goals:** Determine if there is a need to foster innovation within the organization. New technology might be required to support research and development, enable new business models, or drive product innovation.
 2. **Scalability and Future-Proofing:** Assess whether the organization needs to future-proof its operations by adopting scalable and adaptable technology. This ensures that the organization can handle future growth and technological advancements without requiring frequent overhauls.
- vii. **Stakeholder Input:**
 1. **User Feedback:** Gather feedback from end-users, employees, and other stakeholders who interact with the current technology. Identifying pain points, frustrations, or desired features can help justify the need for a new solution.
 2. **Executive Vision:** Align the technology need with the vision and priorities of senior management and other key decision-makers. This ensures that the adoption of new technology is supported at the highest levels of the organization.

viii. Gaps in Current Capabilities:

1. **Capability Assessment:** Conduct an assessment of current systems and technologies to identify gaps or limitations that hinder the organization's ability to achieve its goals. This might involve identifying outdated technologies, unsupported features, or limited scalability.
2. **Benchmarking:** Compare the organization's current technology stack with industry benchmarks or best practices to identify areas where new technology could provide a competitive advantage. Refer Stack OverFlow or any other industry benchmarks.

2. Technology Evaluation Team (TET)

- a) **Form a Technology Evaluation Team:** CTO/Solution Architect to assemble a cross-functional team consisting of members from relevant departments such as Project Management, IT(hardware), software development, testing and cybersecurity to carryout a preliminary research.
- b) CTO/Solution Architect to identify and onboard any consultant/official from industry having worked-on or having knowledge of the relevant technology.

3. Preliminary Research

- a) **Market Analysis:** The Technology Evaluation Team to conduct a market analysis to identify potential technologies that could address the identified need. This includes researching industry trends, competitors' technology stacks, and vendor offerings. Also refer the technology trends by reviewing the analysis reports published by Stack Overflow or Gartner etc.
- b) **Shortlist Technologies:** Based on the market analysis, create a shortlist of 3-5 technologies that appear to be the most promising.

4. Detailed Evaluation

- a) **Technical Feasibility:** TET to evaluate the technical feasibility of each shortlisted technology. Consider factors such as compatibility with existing systems, scalability, performance, security, support forums, and maturity.
 - i. **Compatibility with Existing Systems:**
 1. **Integration:** Assess whether the new technology can be easily integrated with the organization's current software, hardware, and network infrastructure. Consider potential challenges related to data migration, interoperability, and API compatibility.

2. Legacy Systems: Evaluate how the new technology will interact with any legacy systems in place. Determine if the new technology can coexist with these systems or if modifications are necessary.
- ii. Open Source Alternatives: Evaluate if the new technology is open source or any open source alternatives are available with minimal difference in the features.
- iii. Scalability: Determine if the technology can scale to meet future demands. This involves assessing whether the technology can handle an increase in users, data, or transactions without a significant loss in performance or requiring major changes.
- iv. Performance: Analyze the technology's ability to perform efficiently under expected workloads. This includes evaluating response times, processing speeds, and overall system performance.
- v. Security: Assess the technology for potential security vulnerabilities. This includes evaluating the technology's ability to protect data, prevent unauthorized access, and comply with security standards.
- vi. Support:
 1. Vendor Support: Evaluate the level of support provided by the technology vendor, including documentation, customer support, and regular updates.
 2. Forums & Documentation: Evaluate if there are active forums and ample subscribers to support during implementation. Also, check if online documentation is available.
- vii. Technology Maturity: Assess whether the technology has a proven track record of successful implementation in similar environments. Emerging technologies may offer cutting-edge features but could also carry higher risks due to a lack of widespread adoption or unresolved issues.
- viii. Compliance with Standards: Ensure that the technology complies with relevant industry standards, regulations, and best practices. This is particularly important in sectors where compliance is critical, such as government, finance or healthcare etc.
- b) Cost Analysis: Perform a cost-benefit analysis, taking into account the costs associated with acquiring, implementing, maintaining, and training personnel on the new technology. Below are some of the costs which may be considered based on applicability.
 - i. Initial Acquisition Costs: May include Licensing Fees, Hardware Costs, Installation and Setup Costs etc

- ii. **Implementation Costs:** May include Training Costs, External resource cost, Consultation fees, costs of integrating the new technology with existing systems, including any necessary custom development, middleware, or APIs.
 - iii. **Indirect Costs:** May include potential impact of the new technology on employee productivity during the transition period. This could include downtime, learning curves, or temporary reductions in efficiency.
 - iv. **Cost-Benefit Analysis:** If feasible, compare the total costs against the expected benefits of the technology, such as improved efficiency, cost savings, increased revenue, enhanced security, or competitive advantage.
- c) **Resource Availability:** Assess the availability of resources (human, financial, and technological) required to implement and maintain the technology.
- i. **Human Resources:**
 - 1. **Skills and Expertise:** Determine if the current team has the necessary skills and expertise to work with the new technology. If not, consider the time and cost required for training or hiring new staff with the appropriate skills.
 - 2. **Workload Capacity:** Assess whether the existing team has the bandwidth to take on the additional responsibilities that the new technology might bring, without compromising on their current duties.
 - ii. **Financial Resources:** Analyze the available budget to see if it can accommodate the costs associated with adopting the new technology, including licensing fees, hardware upgrades, training, and ongoing maintenance.
 - iii. **Technological Resources:** Evaluate whether the existing technological infrastructure (e.g., servers, networks, databases) is compatible with the new technology or if upgrades will be required.
- d) **Risk Assessment:** Identify and analyze potential risks associated with adopting the new technology. Consider factors such as vendor reliability, technology maturity, and potential for obsolescence.
- e) **Proof of Concept (PoC):** Wherever feasible, develop a PoC to test the technology in a controlled environment. This helps validate the technology's performance, usability, and integration capabilities.

5. Evaluation Report by TET

- a) The TET to prepare a detailed evaluation report as per Annexure-I on the new technology and submit for review to CTO/Solution Architect.

6. Final Decision

- a) Final Review: Review all gathered data and feedback to make an informed decision on the best technology to adopt with CTO.
- b) Approval Process: Obtain necessary approvals from CEO and relevant stakeholders before proceeding with the adoption.
- c) Documentation: Document the entire evaluation process, including the rationale behind the final decision, the technologies considered, and any potential risks identified.

7. Implementation Planning

- a) Implementation Roadmap: Develop a detailed implementation roadmap that outlines the steps, timelines, and resources required for rolling out the new technology.
- b) Training and Support: Plan for necessary training sessions to ensure that all relevant personnel are equipped to use and support the new technology.
- c) Monitoring and Review: Establish monitoring mechanisms to track the implementation progress and address any issues that arise during the adoption phase.

8. Post-Implementation Review

Post implementation after a period of 6-12 months carry out a review to evaluate if everything is going in right direction as planned.

- a) Performance Evaluation: After implementation, conduct a performance evaluation to ensure the new technology meets the organization's needs.
- b) Continuous Improvement: Gather feedback from users and stakeholders to identify areas for improvement and ensure the technology continues to deliver value to the organization.

9. Review and Revision

This SOP should be reviewed and updated annually or whenever there is a significant change in the organization's technology strategy or business goals.

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Annexure-I : Technology Evaluation Report

Problem Statement: _____

Requested/Reported by: _____

Evaluation Period: _____

Technology Evaluation Team: _____

Short-listed options

Detailed Evaluation for all shortlisted options:

1. Compatibility with existing system

2. Open Source alternatives

3. Scalability

4. Performance

5. Security Assessment

6. Support Strategy

7. Maturity

8. Compliance with standards

9. Cost Analysis

a) Initial acquisition cost

i. Hardware

ii. License

iii. Installation/Setup

iv. Network

b) Implementation cost

i. Training

ii. External Consultant/Resources

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iii. Development cost

c) Indirect cost

10. Resource Requirement

a) Human Resources :

i. Existing/Hiring needed

ii. If existing, whether bandwidth available with resource?

iii. Already Trained/Training needed

b) Financial Resources

c) Technological Resources

Risk Assessment: _____

PoC findings: _____

Conclusion: _____

References : _____

CTO/Solution Architect Comments: _____

Sign with Date _____

CEO Comments: _____
